Time series as rough paths

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Stock price models

1. Samuelson stock model- 1965, Hull-White model - 1987:

$$dS_t = \mu_t S_t dt + \sigma_t S_t dB_t, \tag{0.1}$$

   for a stock price S_t, using Itô calculus. σ_t satisfies a stochastic differential equation

$$d \log \sigma_t = k(\theta - \log \sigma_t)dt + \gamma dW_t \tag{0.2}$$

   with parameters k, θ, γ and another Brownian motion W_t, with an instantaneous correlation $dB_t dW_t = \rho dt$ with a parameter $\rho \in [0, 1]$ between the two different Brownian motions

2. Heston model - 1993: σ_t^2 follows a Cox-Ingersoll-Ross equation

$$d(\sigma_t^2) = k\left[\theta - (\sigma_t^2)\right] dt + \lambda \sigma_t dW_t' \tag{0.3}$$

   where $k, \theta, \lambda > 0$ are parameters, and the two Brownian motions again possess an instantaneous correlation $dW_t dW_t' = \rho dt$ with $\rho \in [0, 1]$.

Memory effect

1. Comte-Renault - 1998: ordinary Brownian motion is replaced by fractional Brownian motion in the model (0.2) for the variance σ_t, resulting in

$$d \log \sigma_t = k(\theta - \log \sigma_t)dt + \gamma dB_t^H. \tag{0.4}$$

   In order to obtain a process with long memory, a Hurst exponent $H > \frac{1}{2}$ is needed.

2. Gatheral et al. - (2016,2018): if we assume the model (0.4), the log-volatility behaves essentially as a fractional Brownian motion with Hurst exponent H of order 0.1 at any reasonable time scale. Volatility is rough!

3. Using Hairer's regularity structure theory, Bayer et al. 2019 considers

$$\begin{aligned} \frac{dS_t}{S_t} &= f(Z_t)(\rho dW_t + \sqrt{1-\rho^2} dW_t') \\ Z_t &= z + \int_0^t K(s,t)v(Z_s)ds + \int_0^t K(s,t)u(Z_s)dW_s, \end{aligned} \tag{0.5}$$

4. Our idea: consider first equation, using rough path theory (different rough path lift), not Itô calculus. No arbitrage is proved under transaction cost.

Long memory and multi-scaling phenomenon

	1M	5M	15M	1H	4H	Daily	Weekly
Sp500	0.5158	0.5103	0.5234	0.5201	0.5278	0.5613	0.5816
Dow Jones	0.5199	0.5177	0.5301	0.5437	0.5534	0.5764	0.5952
Nasdaq100	0.5177	0.5193	0.5245	0.5324	0.5490	0.5694	0.6069

Table: Hurst exponents for different time scales using [R/S] analysis.

	1M	5M	15M	1H	4H	Daily	Weekly
Sp500	0.4898	0.4911	0.4969	0.4838	0.4818	0.5649	0.6006
Dow Jones	0.4937	0.4952	0.5021	0.4927	0.4951	0.5049	0.5386
Nasdaq100	0.4907	0.4964	0.5048	0.5031	0.4755	0.5188	0.5651

Table: Hurst exponents for different time scales using Spectral analysis.

	1M	5M	15M	1H	4H	Daily	Weekly
Sp500	0.5152	0.5202	0.5157	0.5181	0.5149	0.5549	0.5410
Dow Jones	0.5201	0.5208	0.5174	0.5161	0.5081	0.5651	0.5734
Nasdaq100	0.5271	0.5294	0.5224	0.5307	0.5288	0.6112	0.6490

Table: Hurst exponents for different time scales using Higuchi method.

Quadratic relation between variance and expectation of the logarithmic return

A drawback of model (0.1) is the fact that

$$\mathbb{E}Y_{t,t+h} = h\Big(\mu - \frac{\sigma^2}{2}\Big), \quad \text{Var } Y_{t,t+h} = \sigma^2 h,$$

which implies that the variance depends linearly and negatively on the expected return

$$\text{Var } Y_{t,t+h} = 2\mu h - 2\mathbb{E}Y_{t,t+h}. \tag{0.6}$$

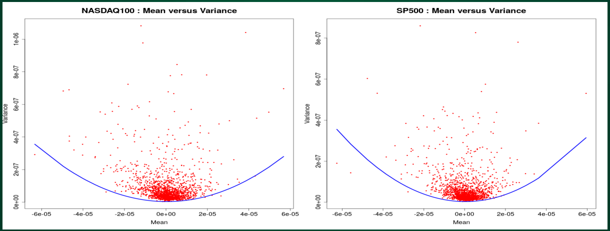


Figure: Upper-parabolic mean-variance relation for 1-minute quotes of stock indices. Data: Dax, Nasdaq100, Nikkei, SP500. Data source: Dukascopy Bank SA

Rough model for asset price

Theorem [LHD-JJost (SIFIN-2023)]: Consider $dS = \mu_t S_t dt + \sigma_t S_t dX_t$. Then $S_t = e^{Y_t}$ where

$$Y_b = Y_a + \int_a^b \mu_u du - \frac{1}{2}\int_a^b \sigma_u^2 d[x]_u + \int_a^b \sigma_u dx_u \tag{0.7}$$

The first integral is of classical Riemann type, the second is of Young type for $\sigma^2 \in C^\beta$ with respect to the path $[x] \in C^{2\alpha}$. The third one is a rough integral for σ controlled by $(\omega, x)^{\mathrm{T}}$.

Non-trivial example: When $X = \hat{B}$ is the G-Brownian motion with respect to the sublinear expectation $\mathbb{E}$, then it follows from Shige Peng that $[\hat{B}] = \langle \hat{B} \rangle$ and one can solve the stochastic differential equation

$$S_t = S_0 + \int_0^t \mu S_u du + \int_0^t \sigma S_u d\hat{B}_u \tag{0.8}$$

explicitly as

$$Y_{s,t} = \mu(t-s) + \sigma \hat{B}_{s,t} - \frac{1}{2}\sigma^2 \Big(\langle \hat{B} \rangle_t - \langle \hat{B} \rangle_s\Big). \tag{0.9}$$

Subordinate stochastic process of the form $B^H(T_t)$ where T is a multi-fractal stochastic process, which according to Mandelbrot & Taylor (1967) might relate to the local transaction time, while according to Clark (1973) might be the trading volume.

Gradient decent method

- Minimization problem

$$\min_{\theta\in\mathbb{R}^d} Q(\theta), \quad Q \in C^1. \tag{0.10}$$

- Global solution: $\theta^* \in \arg\min_{\theta\in\mathbb{R}^d} Q(\theta) \Rightarrow \nabla Q(\theta^*) = 0$. This leads to an approximation scheme to achieve θ^*, which is the zero solution of $\nabla Q(\theta) = 0$

$$\theta_{t+1} = \theta_t - \eta \nabla Q(\theta_t), \tag{0.11}$$

for the learning rate $\eta > 0$. (0.11) is called Gradient descent method.
- When η small enough, (0.11) can also be viewed as the Euler scheme for solving the ODE

$$\dot{\theta} = -\nabla Q(\theta), \tag{0.12}$$

where one attempts to solve for the steady state θ^* of (0.12).

Stochastic gradient descent

- Minimization problem

$$\min_{\theta\in\mathbb{R}^d} Q(\theta) = \min_{\theta\in\mathbb{R}^d} \frac{1}{m}\sum_{i=1}^{m} Q^{(i)}(\theta), \quad Q^{(i)} \in C^1. \tag{0.13}$$

  When $m \gg 1$, (0.11) is replaced by: at each step t choose a random set Ω_t of $\{1,\dots,m\}$ (with or without replacement) and then updating

$$\theta_{t+1} := \theta_t - \eta\frac{1}{|\Omega_t|}\sum_{i\in\Omega_t} \nabla Q^{(i)}(\theta_t) = \theta_t - \eta\nabla\tilde{Q}_t(\theta_t) \tag{0.14}$$

  No matter how random the choice, can rewrite SGD recursion

$$\theta_{t+1} = \theta_t - \eta\nabla Q(\theta_t) + \eta^{\alpha}\epsilon_t(\theta_t). \tag{0.15}$$

- (0.15) can also be viewed as the Euler scheme of the "limiting equation"

$$d\theta = -\nabla Q(\theta)dt + \sigma(\theta)dX_t, \tag{0.16}$$

  where $\sigma\sigma^{\mathrm{T}} = \eta^{2\alpha}\mathbb{E}[\epsilon\epsilon^{\mathrm{T}}|\theta_t = \theta]$.
- If Ω_t i.i.d. then CLT gives $\alpha = \frac{1}{2}$ and X-Gaussian. But not necessary i.i.d.
- Questions: How does the distribution of (0.15) converge to some limit? And which calculus is used to interpret and solve the limiting equation?

Signatures of rough paths

Let $x \in C^{1-var}(I, \mathbb{R}^d)$. x_t^i is the coordinate i of x_t at time t. Define

$$\mathbf{g}^{k;i_1,\cdots,i_k} := \int_s^t \int_s^{u_k} \ldots \int_s^{u_2} dx_{u_1}^{i_1} \ldots dx_{u_k}^{i_k}.$$

and the set

$$\mathbf{g} = \left(\mathbf{g}^{k;i_1,\cdots,i_k} : 1 \leq k \leq N; i_1, \ldots, i_k \in \{1, \ldots, d\} \right)$$

to be the N-level signature of $x|_{[s,t]}$ and is written as $S_N(x)_{s,t}$. In other words,

$$S_N(x)_{s,t} \equiv \left(1, \int_{s<u<t} dx_u, \ldots, \int_{s<u_1<\ldots<u_k<t} dx_{u_1} \otimes \ldots \otimes dx_{u_k} \right)$$ Then

$$\in \oplus_{k=0}^N \left(\mathbb{R}^d \right)^{\otimes k},$$

$S_N(x)_{s,\cdot}$ satisfies the controlled DE by x as follows

$$\begin{cases} dS_N(x)_{s,t} = S_N(x)_{s,t} \otimes dx_t, \\ S_N(x)_{s,s} = 1. \end{cases}$$

Rough differential equation

$$dy = V(y)dx \Rightarrow y_{s,t} \approx \tilde{V}(y_s) \otimes S_N(x)_{s,t}$$

Fitzhugh-Nagumo neuro-system

The deterministic FHN neuron model is

$$\begin{cases} dv_t &= v_t - \frac{v_t^3}{3} - w_t + I \\ dw_t &= \varepsilon(v_t - \mu w_t + J) \end{cases} \tag{0.17}$$

Perturbed by bounded noise

$$dy = f(y)dt + h(y)dB^H \tag{0.18}$$

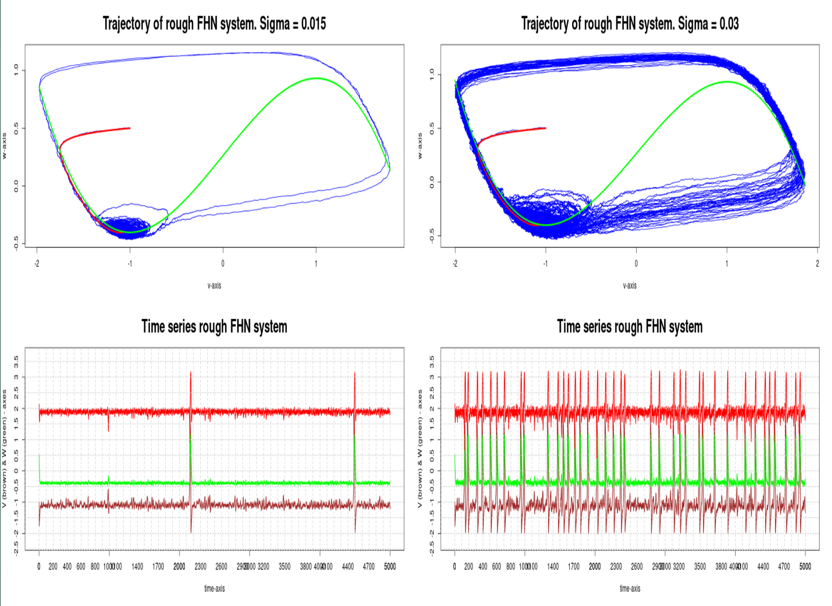
for a certain constant C_f. Instead, we introduce a different Lyapunov function

$$V(y) := \Big(1 + v^4 + \frac{6}{\varepsilon\mu} w^2\Big)^{\frac{1}{4}}. \tag{0.19}$$

Theorem [LHD-J.Jost (2023)]: Under the assumptions $(\mathbf{H}_g^b)$, $(\mathbf{H}_X)$, there exists a solution of stochastic FHN system on any interval $[0,T]$. Moreover, there exist constants $\delta > 0$ such that for any $\lambda \in (0,1)$ small enough, there exist a constant $K_\lambda > 0$ such that the following estimate holds for all

$$V(y_t) - K_\lambda \leq e^{-\delta t}\Big[V(y_0) - K_\lambda\Big] + L_V \lambda \int_0^t e^{-\delta s} dN^-\Big(\frac{\lambda}{16 C_p C_g}, \mathbf{x}, [t-s, t]\Big), \quad \forall t \in [0,T]. \tag{0.20}$$

Consequently, there exists a global pullback attractor $\mathcal{A}$.



Trajectory of rough FHN system. Sigma = 0.015
w-axis
v-axis
Trajectory of rough FHN system. Sigma = 0.03
w-axis
v-axis
Time series rough FHN system
V (brown) & W (green) - axes
time-axis
Time series rough FHN system
V (brown) & W (green) - axes
time-axis